7.2 Population dynamics and "Critical Catastrophe" (5/20 pts) – estimated 45min

General information. This is an example from the Exam of 2024, based mainly on understanding Poisson processes, and how to write down a Master equation, and ensemble averages. The exam has exercises of various difficulty levels, this one is considered a medium difficulty level.

Evaluated here: Poisson processes (advanced), Markov chains and writing a master equation, definition of ensemble averages, studying first and second order moments, computing their equation from the master equation, solving simple dynamics, reflecting on the results.

Birth-Death process (Galton-Watson process)

We study the evolution of a population of individuals who can reproduce and die.

For each individual, birth or death events happen randomly, and follow a Poisson process with rate λ. An event is then either a death with probability p^-, or a birth with probability p^+, such that $p^- + p^+ = 1$.

When a birth happens, the number of individuals in the population is increased by 1. When an individual dies, it is removed from the population, and therefore the number of individuals in the population is reduced by 1.

At time $t = 0$, there are N_0 individuals in the population. We are interested in studying how the number $N(t)$ of individuals in the population evolves in time.

Q1. What is (or are) the absorbing state(s) of the system?

Q2. We denote ν the mean number of individuals created when an event (birth or death) occurs. Can you show that:

$$\nu = p^+ - p^- \quad ? \tag{7.7}$$

We will see that the evolution of the population depends on the value of ν.

Q3. a. Take one individual in the population. What is the rate at which it undergoes an event regardless of its type (birth or death)? What is the rate at which it gives birth? What is the rate at which it may die?

b. Take a population of N individuals. What is the rate at which an event (birth or death) occurs in the whole population? What is the rate at which a birth occurs in the population? What is the rate at which a death occurs in the population?

We want to study the evolution of the number $N(t)$ of individuals in the population in time. As the birth-death process is random, $N(t)$ is a random variable. For instance, if one were to simulate the evolution of the population, each simulation would give a different result.

Q4. To study the evolution of the population size, the most general approach consists in studying the evolution of the probability $P(N,t)$ that there are N individuals in the population at time t. Can you write down what is the probability $P(N, t+dt)$ to find N individuals in the population at time $t+dt$ as a function of the state of the system at time t, and show that $P(N,t)$ satisfies the master equation:

$$\frac{\partial P(N,t)}{\partial t} = -\lambda\, N\, P(N,t) + \lambda\, p^+\, (N-1)\, P(N-1,t) + \lambda\, p^-\, (N+1)\, P(N+1,t) \ ? \tag{7.8}$$

Please explain each term in this equation.

This equation is difficult to solve due to its recursive structure: to compute $P(N,t)$ we must know $P(N+1,t)$ and $P(N-1,t)$; to compute $P(N+1,t)$ we must know $P(N+2,t)$ and $P(N,t)$, etc. An alternative is to study the evolution of the moments of $N(t)$, instead of $P(N,t)$. Below, we focus on the first two moments.

The *Critical catastrophe*

From the master equation (7.16), one can compute the equation of evolution of any moments of N. Here we are only interested in the time evolution of the mean $\langle N\rangle(t)$ and of the variance $\sigma^2(t) = \langle N^2\rangle(t) - \langle N\rangle^2(t)$, where the brackets $\langle\cdot\rangle$ denote the ensemble average.

Q5. Fixing a time t, what is the definition of the ensemble averages $\langle N(t)\rangle$ and $\langle N^2(t)\rangle$?

(⋆) Q6. Can you prove the two following relations?

$$\sum_{N\geq 1} N(N-1)\,P(N-1) = \left\langle N^2\right\rangle + \langle N\rangle \qquad \text{and} \qquad \sum_{N\geq 0} N(N+1)\,P(N+1) = \left\langle N^2\right\rangle - \langle N\rangle\ . \tag{7.9}$$

Note: these relations are valid at any time t. here, for clarity, we removed the dependence in t, i.e. we wrote $P(N)$ instead of $P(N,t)$, and N instead of $N(t)$. You can also temporarily remove the dependence on time in your answer.

Q7. Using these relations and the master equation (7.16), can you show that $\langle N\rangle(t)$ follows the equation:

$$\frac{d\,\langle N\rangle(t)}{dt} = \lambda\,\nu\,\langle N\rangle(t)\ ? \tag{7.10}$$

Q8. What is the solution of this equation? We recall that at time $t = 0$, there are exactly N_0 individuals in the population, and therefore $\langle N\rangle(t=0) = N_0$. The system is said to be *critical* for $\nu = 0$, *super-critical* for $\nu > 0$, and *sub-critical* for $\nu < 0$. What is the behavior of $\langle N\rangle(t)$ in these three regimes? How is the number of individual in the population expected to be after a very long time?

(⋆) **Q9.** Let us now focus on the variance of $N(t)$. Using the two following relations:

$$\sum_{N\geq 1} N^2(N-1)P(N-1) = \left\langle N^3\right\rangle + 2\left\langle N^2\right\rangle + \langle N\rangle \qquad \text{and} \qquad \sum_{N\geq 0} N^2(N+1)P(N+1) = \left\langle N^3\right\rangle - 2\left\langle N^2\right\rangle + \langle N\rangle \tag{7.11}$$

and the master equation (7.16), can you show that $\langle N^2\rangle(t)$ follows the equation:

$$\frac{d\,\langle N^2\rangle(t)}{dt} = 2\,\lambda\,\nu\,\langle N^2\rangle(t) + \lambda\,\langle N\rangle(t)\ ? \tag{7.12}$$

(⋆) **Q10.** Can you show that the variance $\sigma^2 = \left\langle N^2\right\rangle - \langle N\rangle^2$ follows the same equation:

$$\frac{d\,\sigma^2(t)}{dt} = 2\,\lambda\,\nu\,\sigma^2(t) + \lambda\,\langle N\rangle(t)\ ? \tag{7.13}$$

Q11. In the critical case $\nu = 0$**,** Eq. (7.13) becomes:

$$\frac{d\,\sigma^2(t)}{dt} = \lambda\langle N\rangle(t)\,, \qquad \text{where} \qquad \langle N\rangle(t) = N_0 \qquad \text{is a constant}\,. \tag{7.14}$$

Can you solve this equation to find $\sigma^2(t)$? We recall that at time $t = 0$, there are exactly N_0 individuals in the population, and therefore $\sigma^2(t=0) = 0$.

Q12. Critical catastrophe. In the critical case $\nu = 0$:

a. What is the behavior of $\sigma(t)$ at large times? We would like to compare this behavior to the behavior of $\langle N\rangle(t)$ at large times. Can you compute the ratio $\sigma(t)/\langle N\rangle(t)$?

b. Observe that there is a characteristic time τ_c at which the standard deviation $\sigma(t)$ becomes larger than mean $\langle N\rangle(t)$. What is this characteristic time? What happens around that time? This phenomenon is known as the *critical catastrophe*. Can you comment on it? What do you think of the common approach, which consists in studying only the evolution of the mean number of individual $\langle N\rangle(t)$ in the context of population dynamics? Is it always a good idea?

Hints and intermediate solutions (not given in the exam)

For Q4. To study the evolution of the population size, the most general approach consists in studying the evolution of the probability $P(N,t)$ that there are N individuals in the population at time t. Can you show that the probability to find N individuals in the population at time $t+dt$ is given by:

$$P(N,t+dt) = P(N,t) \times [1 - N\,\lambda\,dt] + P(N-1,t) \times (N-1) \times \lambda\,p^{+}\,dt + P(N+1,t) \times (N+1) \times \lambda\,p^{-}\,dt \ ? \tag{7.15}$$

Please explain each term in this equation. Can you then show that $P(N,t)$ verifies the master equation:

$$\frac{\partial P(N,t)}{\partial t} = -\lambda\,N\,P(N,t) + \lambda\,p^{+}\,(N-1)\,P(N-1,t) + \lambda\,p^{-}\,(N+1)\,P(N+1,t) \ ? \tag{7.16}$$

Additional questions (which were given in a longer version of the exercise)

Q9.bis. Can you prove the two following relations?

$$\sum_{N\geq 1} N^2(N-1)P(N-1) = \left\langle N^3 \right\rangle + 2\left\langle N^2 \right\rangle + \langle N \rangle \ , \tag{7.17}$$

and

$$\sum_{N\geq 0} N^2(N+1)P(N+1) = \left\langle N^3 \right\rangle - 2\left\langle N^2 \right\rangle + \langle N \rangle \ . \tag{7.18}$$

Q13. (⋆⋆) a. In the case $\nu \neq 0$, the solution of Eq. (7.13) takes the form:

$$\sigma^2(t) = A(t)\,\mathrm{e}^{2\lambda\nu t}\,, \tag{7.19}$$

where $A(t)$ is a function of time t. Replacing this solution in Eq. (7.13), can you compute $A(t)$ and show that:

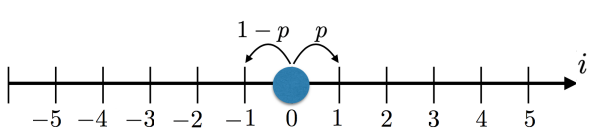
$$\sigma^2(t) = \frac{N_0}{\nu}\left(\mathrm{e}^{2\lambda\nu t} - \mathrm{e}^{\lambda\nu t}\right) \ ? \tag{7.20}$$

We recall that at time $t=0$, there are exactly N_0 individuals in the population, and therefore $\sigma^2(t=0)=0$.
b. In the sub-critical case $\nu < 0$, what is the behavior of $\sigma^2(t)$ at large time t?
c. In the super-critical case $\nu > 0$, what is the behavior of $\sigma^2(t)$ at large time t? What is the behavior of the ratio $\sigma/\langle N\rangle(t)$ at large time t? Can you comment on this result?

7.3 Random walks and control theory (5/20 pts) – estimated 0h45

General information. This is an example from the Exam of 2023. The topics assessed here are: Random walks, summing of i.i.d random variables, Markov chains and master equation, Diffusion equation, simulation of random walks, critical assessment of the results. The exam has exercises of various difficulty levels, this one is considered to go from easy to medium to high difficulty level, depending on the question (see ⋆ indications).

This exercise is inspired from the paper *Effective Langevin equations for constrained stochastic processes* by Satya Majumdar and Henri Orland (2015).



Q1. 1D discrete Random Walk. Consider a random walker evolving on a one dimensional lattice (see Figure above): at each time step the walker jumps to the next site to the right or to the left with the same probability $p = 1/2 = (1-p)$.